

RETAIL INTELLIGENCE CHALLENGE

Smart Inventory & Replenishment System for Retail POS

Build an ML-driven inventory intelligence system that helps retailers predict stockout risk, forecast future demand, and make smarter replenishment decisions.

Problem Statement

Background

Retail businesses frequently face two critical inventory challenges: **stockouts**, which result in lost sales and poor customer experience, and **overstocking**, which increases inventory holding costs and the risk of dead stock.

Modern Retail POS systems continuously generate valuable data through sales transactions, inventory movements, product catalogs, pricing, promotions, suppliers, and store operations. However, this data is often used primarily for historical reporting rather than predictive decision-making.

The objective of this challenge is to develop an **inventory intelligence system** that can analyze historical retail data and provide actionable recommendations to store operators.

1. Stockout Risk Prediction - Classification

Develop a classification model that predicts whether a product is likely to experience a **stockout during an upcoming prediction period**. The model should estimate the probability of stockout and classify each product into:

- **0 - Low Stockout Risk**
- **1 - High Stockout Risk**

2. Future Demand Prediction - Regression

Develop a regression model that predicts the **expected quantity of a product that will be sold during the upcoming prediction period**. For example, a product currently has 15 units in stock while the model predicts demand of 42 units over the next 7 days.

3. Intelligent Replenishment Recommendation

Combine the outputs of the classification and regression models with the current inventory position to generate an actionable replenishment recommendation.

The system should help answer: **Which products are at risk of running out, how much demand should be expected, and how much inventory should potentially be replenished?**

The solution should be designed as a practical ML component that could eventually be integrated into a **Retail POS SaaS platform** and used by store owners, inventory managers, and purchasing teams.

Expected Outcome

The developed solution should produce an inventory intelligence output for each relevant product/store combination containing information such as:

Product	Current Stock	Predicted Demand	Stockout Probability	Risk	Recommended Qty.
Basmati Rice 5kg	12	35	0.87	High	30
Tata Salt 1kg	42	15	0.21	Low	0
LED Bulb 9W	8	27	0.76	High	25

- A classification model for stockout risk prediction.
- A regression model for future demand prediction.
- Predicted stockout probability for each product.

- Predicted future demand quantity.
- Identification of products requiring attention.
- A recommended replenishment quantity or action.
- A clear and explainable methodology for generating the recommendation.
- A reproducible machine learning workflow suitable for integration into a Retail POS SaaS environment.

What the Dataset Must Contain

The dataset should represent realistic **historical retail POS and inventory activity** and contain sufficient information to support both classification and regression tasks.

Product Information

- `product\_id`
- `product\_name`
- `category`
- `subcategory`
- `brand`
- `unit\_of\_measure`
- `supplier\_id`

Store Information

- `store\_id`
- `store\_type`
- `store\_location`
- `city`
- `region`

Historical Sales

The dataset should contain historical sales information at a suitable time granularity, preferably **daily product-store level**, including:

- `date`
- `quantity\_sold`
- `sales\_amount`
- `number\_of\_transactions`
- `average\_selling\_price`

Sufficient historical records should be available to derive rolling and lag-based features such as:

- Previous day sales
- 7-day sales
- 14-day sales
- 30-day sales
- Average daily sales
- Sales trend
- Demand variability

Inventory Information

- `opening\_stock`
- `closing\_stock`

- `current\_stock`
- `stock\_received`
- `stock\_adjustment`
- `stockout\_indicator`
- `previous\_stockout\_count`

Pricing & Promotion

- `selling\_price`
- `purchase\_price`
- `discount\_percentage`
- `promotion\_active`
- `promotion\_type`

Supplier / Procurement Information

- `supplier\_id`
- `supplier\_lead\_time`
- `minimum\_order\_quantity`
- `last\_purchase\_quantity`
- `days\_since\_last\_restock`

Temporal Information

- Day of week
- Week
- Month
- Seasonality
- Holidays or special events where applicable

Target Variables

The dataset must provide or allow the candidate to derive two clearly defined targets:

Classification Target - `stockout\_target`

Indicates whether the product experiences a stockout during the defined future prediction window.

- 0 - No Stockout
- 1 - Stockout

Regression Target - `future\_demand`

Represents the quantity expected to be sold during the defined future prediction window. For example:
future_demand = total quantity sold during the next 7 days.

The prediction horizon should be clearly defined and consistently applied across the dataset.

Data Quality Characteristics

To make the challenge representative of a real-world POS environment, the dataset should contain a realistic combination of:

- Numerical features
- Categorical features
- Missing values
- Different numerical scales
- Outliers
- Imbalanced stockout classes
- Multiple products
- Multiple stores
- Seasonal demand patterns
- Products with different sales velocities

The dataset should contain enough historical records to allow candidates to perform meaningful **train/test splitting, cross-validation, feature engineering, classification, and regression** without making the task excessively large for the expected four-hour assessment.